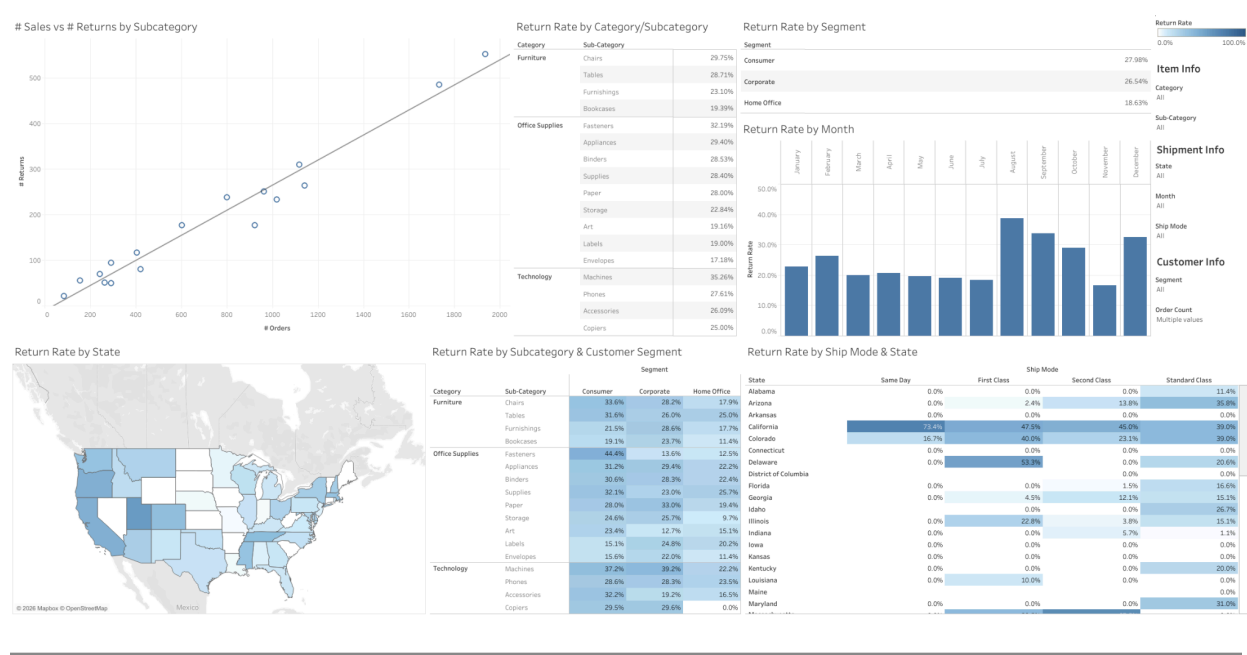


Intro:

Our goal in this analysis is to understand why product returns occur, where they concentrate, and which customer groups or product types are most affected. To do that, we'll use the return rate as our primary measure (ratio of returned orders to total orders).

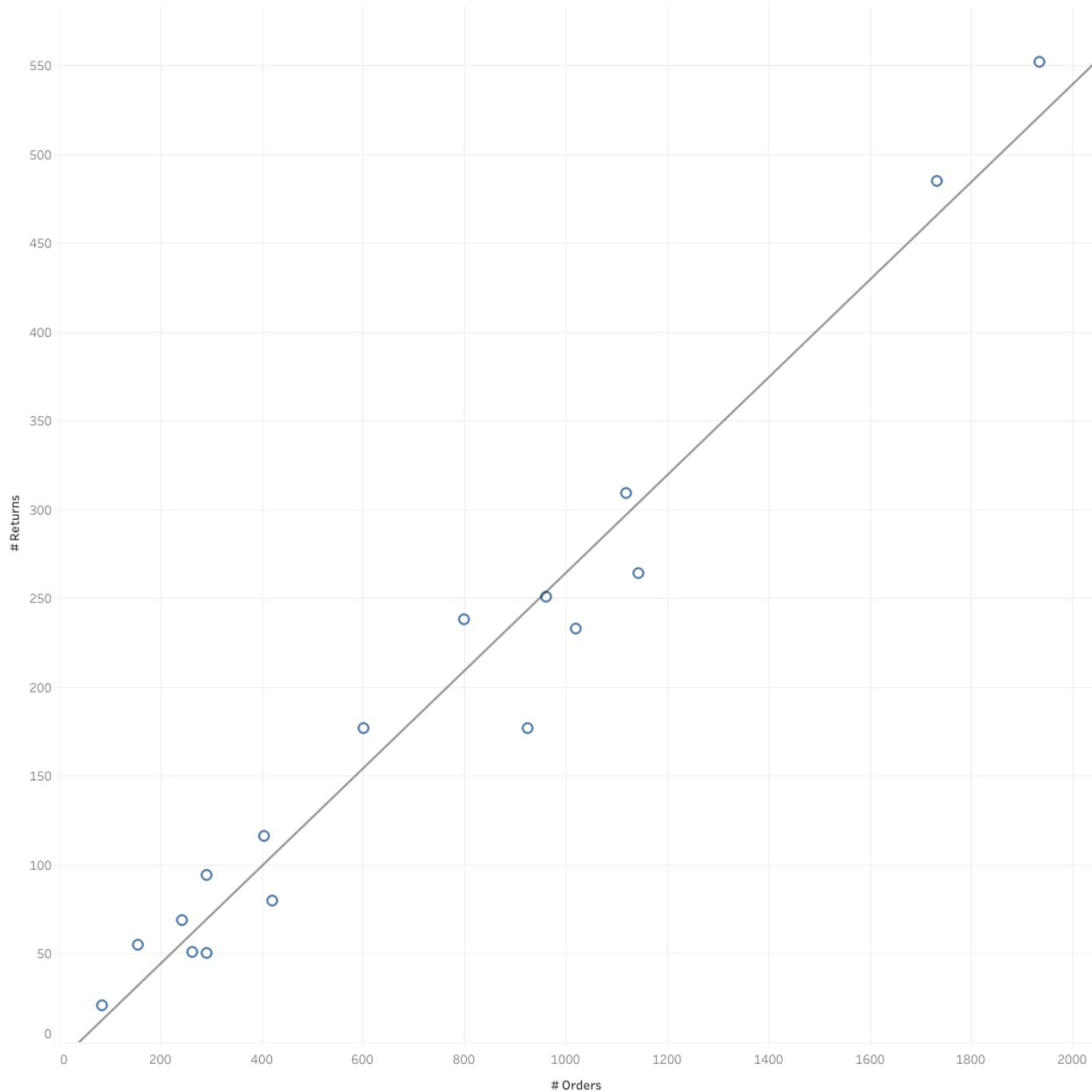
The return rate is the most meaningful metric because it normalizes for sales volume. High-selling categories naturally have more returns, so counts alone can be misleading. This metric lets us compare categories, segments, and regions on equal footing and identify true problem areas.



Graphic #1: Sales vs. Returns by Subcategory

- Most subcategories follow a proportional relationship: more sales -> more returns.
- Points below the trend line indicate better-than-expected performance (fewer returns than predicted).
- Points above the line indicate worse-than-expected performance (more returns than predicted).
- The subcategories above the line are prime candidates for investigation.

Sales vs # Returns by Subcategory



Graphics #2: Return Rate by Subcategory

- Return rates vary significantly within each category.
- Best performers (lowest return rates):
 - Furniture: Bookcases
 - Office Supplies: Envelopes
 - Technology: Copiers
- Worst performers (highest return rates):
 - Furniture: Chairs
 - Office Supplies: Fasteners
 - Technology: Machines
- High return rates may not reflect product quality; they could point to unclear or inaccurate product descriptions, especially for small, inexpensive items.

Return Rate by Category/Subcategory

Category	Sub-Category	
Furniture	Chairs	29.75%
	Tables	28.71%
	Furnishings	23.10%
	Bookcases	19.39%
Office Supplies	Fasteners	32.19%
	Appliances	29.40%
	Binders	28.53%
	Supplies	28.40%
	Paper	28.00%
	Storage	22.84%
	Art	19.16%
	Labels	19.00%
	Envelopes	17.18%
Technology	Machines	35.26%
	Phones	27.61%
	Accessories	26.09%
	Copiers	25.00%

Graphics #3: Return Rate by Segment

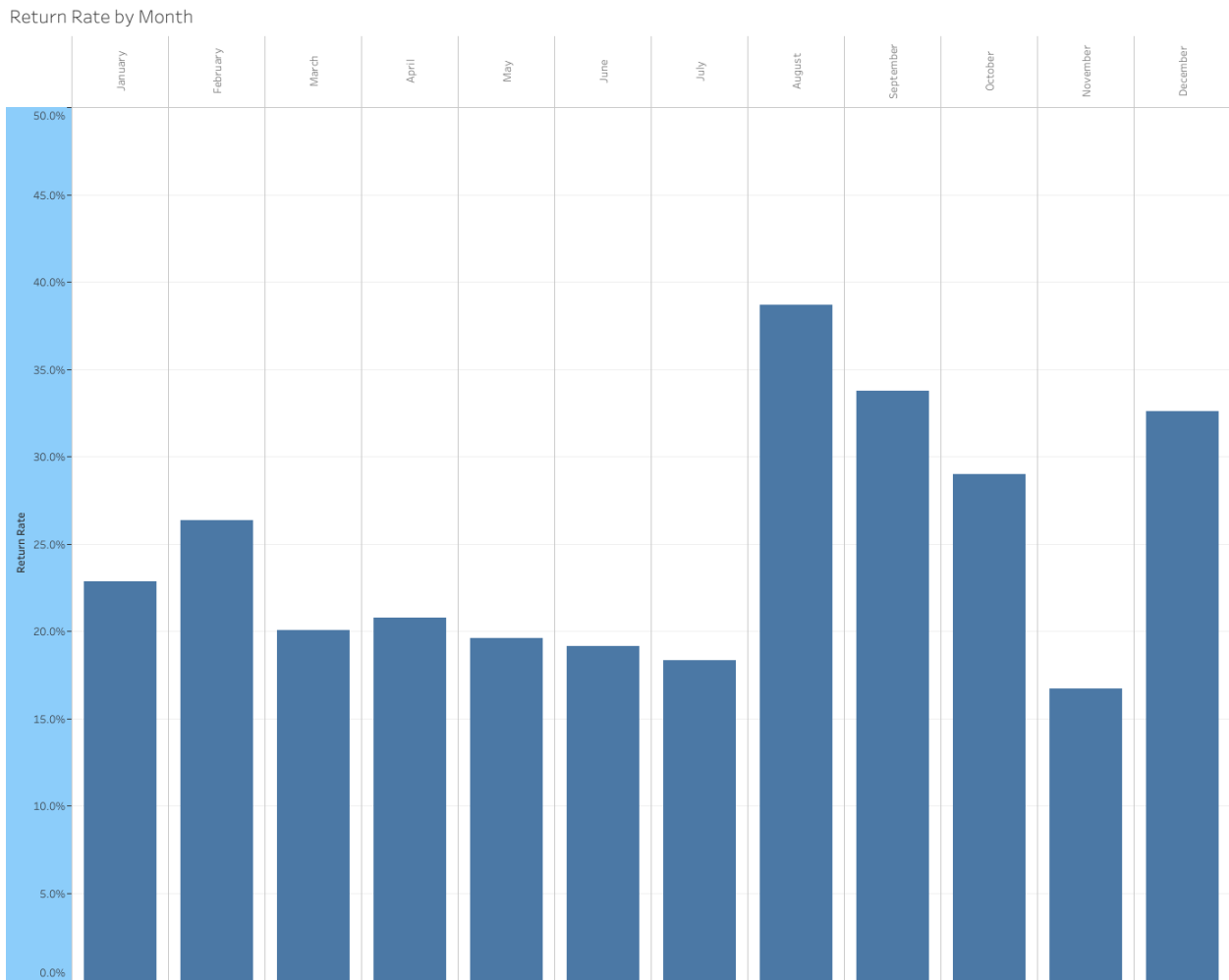
- Consumers have the highest return rate.
- Corporate customers follow closely behind.
- Home Office customers return significantly less.
- This pattern likely reflects predictability:
 - Home offices buy for one person or a small group, leading to fewer surprises & returns
 - Corporate and Consumer purchases involve more people and more variability.

Return Rate by Segment

Segment	
Consumer	27.98%
Corporate	26.54%
Home Office	18.63%

Graphic #4: Return Rate by Month

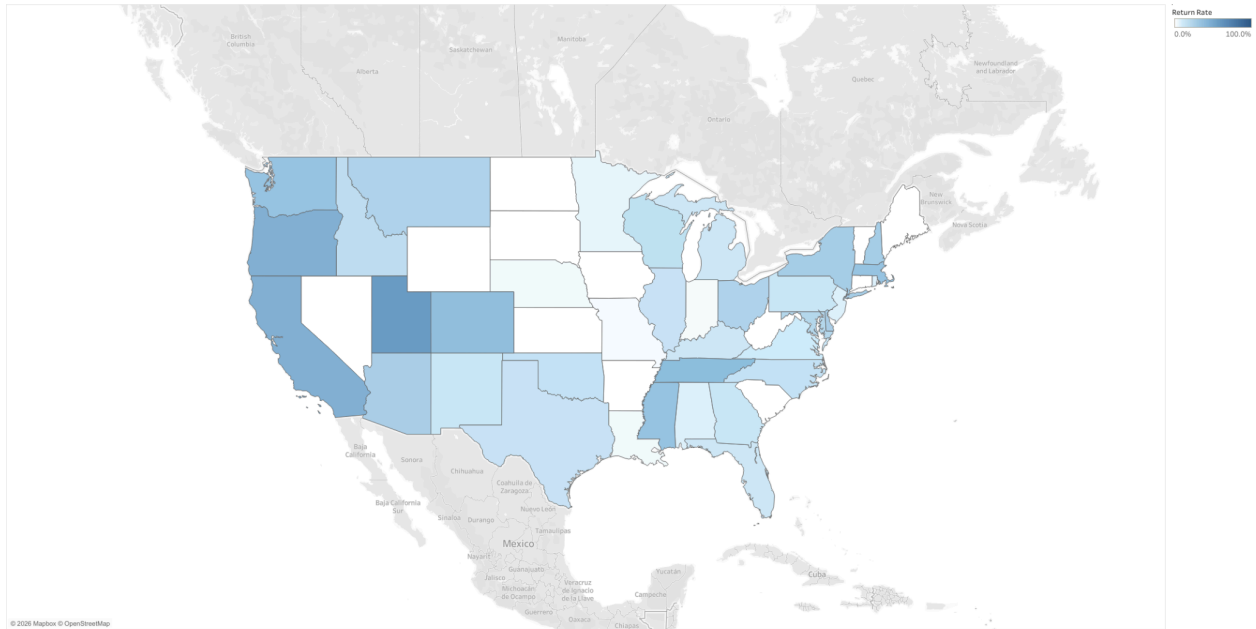
- Returns spike in August, September, October, and December.
- These months align with fiscal year reporting cycles and end-of-year purchasing, when organizations often buy in bulk and adjust orders afterward.



Graphic #5: Return Rate by State

- Certain states show consistently higher return rates, suggesting regional delivery or fulfillment issues.
- These geographic patterns help pinpoint where operational improvements may be needed.

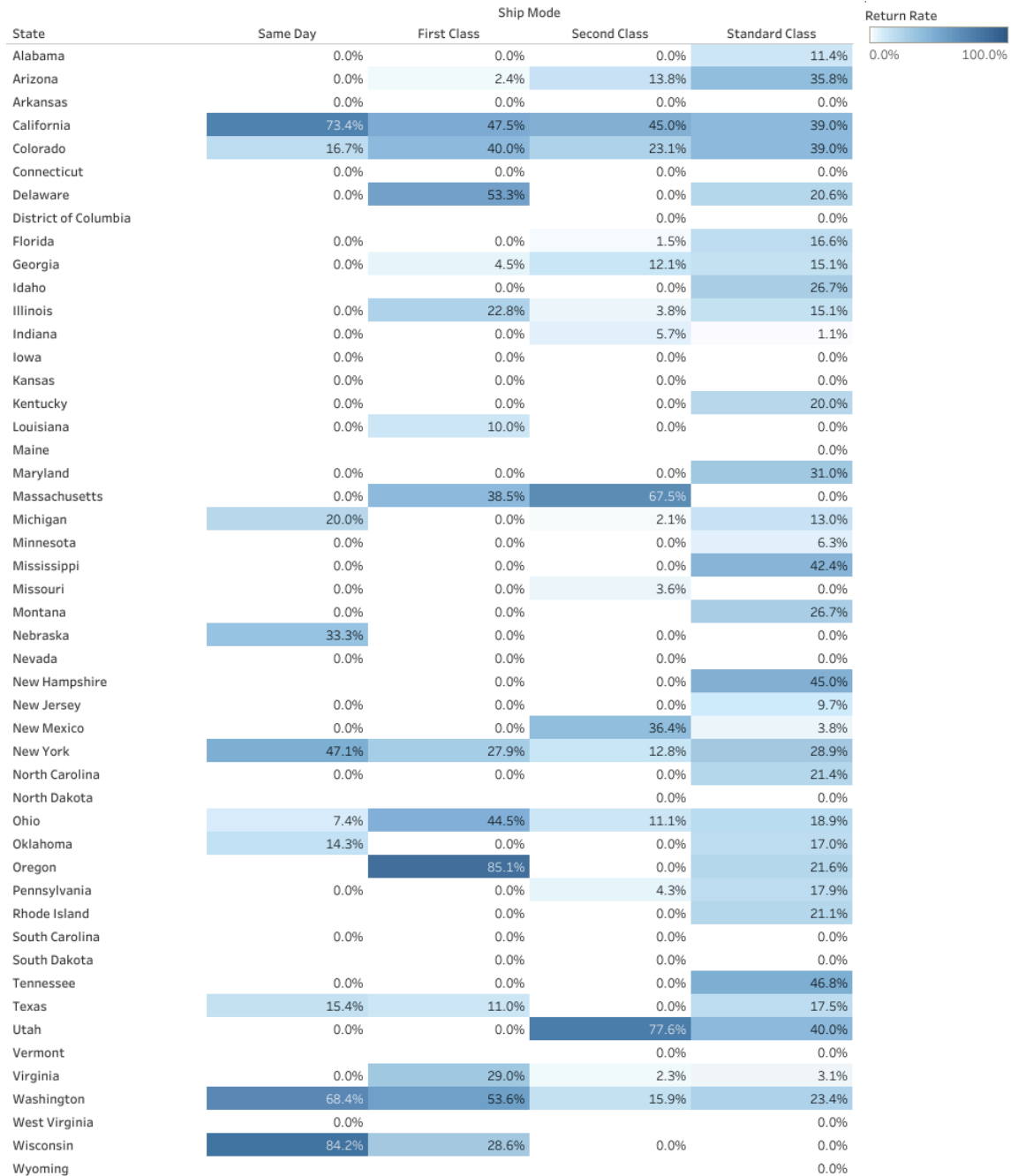
Return Rate by State



Graphic #6: Return Rate by Ship Mode & State

- Standard Class shows widespread issues across many states.
- Some states also struggle with First Class, indicating logistics or carrier-specific problems.
- This view helps isolate whether the issue is state-specific, carrier-specific, or both.

Return Rate by Ship Mode & State



Graphic #7: Return Rate by Subcategory & Customer Segment

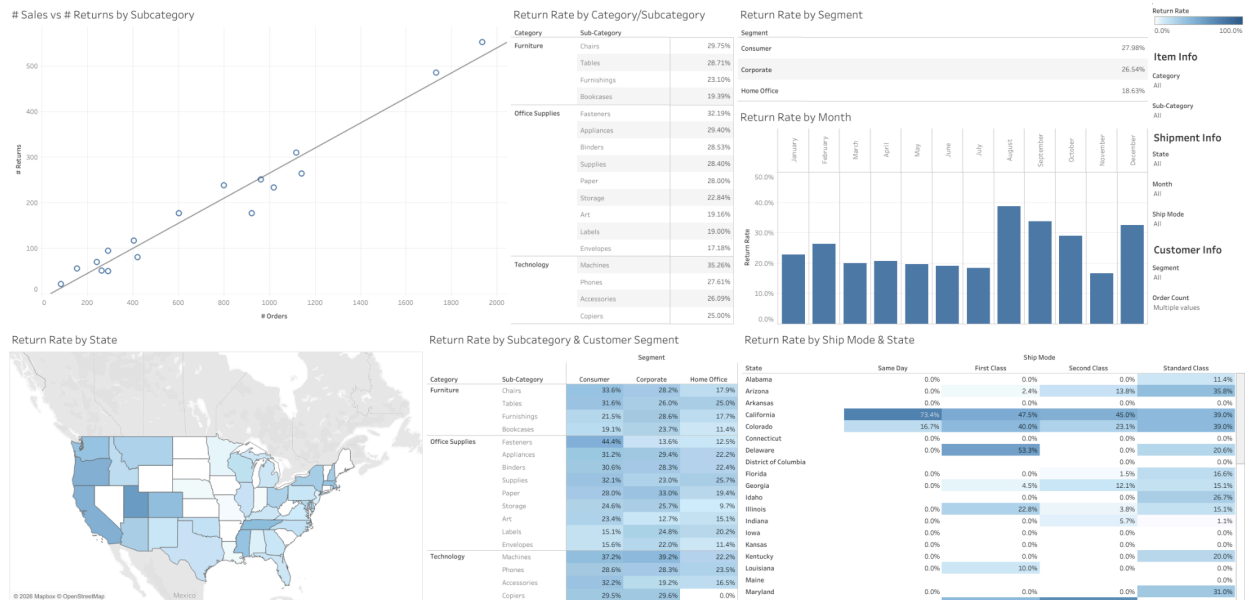
- High return rates cluster around Consumer and Corporate segments for several subcategories.
- This reinforces the idea that product clarity and expectation-setting matter most for these groups.
- Tailoring product descriptions or guidance for these segments could significantly reduce returns.

Return Rate by Subcategory & Customer Segment

Category	Sub-Category	Segment			Avg. Returned?
		Consumer	Corporate	Home Office	
Furniture	Chairs	33.6%	28.2%	17.9%	0.0% 100.0%
	Tables	31.6%	26.0%	25.0%	
	Furnishings	21.5%	28.6%	17.7%	
	Bookcases	19.1%	23.7%	11.4%	
Office Supplies	Fasteners	44.4%	13.6%	12.5%	
	Appliances	31.2%	29.4%	22.2%	
	Binders	30.6%	28.3%	22.4%	
	Supplies	32.1%	23.0%	25.7%	
	Paper	28.0%	33.0%	19.4%	
	Storage	24.6%	25.7%	9.7%	
	Art	23.4%	12.7%	15.1%	
	Labels	15.1%	24.8%	20.2%	
Technology	Envelopes	15.6%	22.0%	11.4%	
	Machines	37.2%	39.2%	22.2%	
	Phones	28.6%	28.3%	23.5%	
	Accessories	32.2%	19.2%	16.5%	
	Copiers	29.5%	29.6%	0.0%	

Dashboard Analysis:

- How to use effectively?
 - a. Start with the scatter plot to identify subcategories that deviate from expected behavior.
 - b. Use the Return Rate tables to drill into which categories or segments are driving issues.
 - c. Filter by Category, Subcategory, Segment, Month, or Ship Mode to isolate patterns.
 - d. Check the map to see whether geographic clusters exist.
 - e. Use the Ship Mode & State table to validate whether delivery method is contributing to the problem.
 - f. Combine filters (e.g., Subcategory + Segment + State) to pinpoint root causes.
- What to monitor?
 - a. Track whether interventions reduce return rates.
 - b. Identify emerging problem areas early.
 - c. Support data-driven conversations with suppliers, logistics partners, and customer service teams.



Conclusions:

- Return behavior is influenced by product type, customer segment, geography, and shipping method.
- High return rates in small items (Office Supplies, Technology accessories) suggest expectation mismatch, not necessarily product defects.
- Consumer and Corporate segments drive most returns, while Home Office is stable.
- Certain states and shipping modes show systemic issues that warrant operational review.

Recommendations:

- Improve Product Descriptions
 - Focus on high-return subcategories like Chairs, Fasteners, and Machines.
 - Add clearer specs, sizing guides, photos, and “what to expect” notes.
 - Tailor descriptions to Consumer and Corporate segments, who return the most.
- Strengthen Quality Control for Delivery
 - Investigate Standard Class shipping issues across states.
 - For states with First Class problems, review carrier performance and packaging standards.
 - Consider regional carrier adjustments or improved packaging for fragile items.
- Segment-Specific Interventions
 - Provide purchase guidance for high-return Consumer and Corporate subcategories.
 - Offer bulk-order previews, sample kits, or “frequently returned due to...” warnings.
- Geographic Focus
 - Target states with consistently high return rates for logistics audits or customer service follow-ups.
 - Use the dashboard to monitor whether interventions reduce return rates over time.